

Graded Homework #11

John Doll

6.2 c. $z = y^2$ $\sqrt{z} = y$ $\frac{dy}{dz} = \frac{1}{2\sqrt{z}}$

$$F(z) = \left(\frac{1}{2\sqrt{z}}\right) f(y) = \left(\frac{1}{2\sqrt{z}}\right) \left(\frac{3z}{2}\right) = \frac{3\sqrt{z}}{4}$$

$$F(z) = f(\sqrt{z}) + f(-\sqrt{z}) = \frac{3\sqrt{z}}{2}$$

$$f(z) = \begin{cases} \frac{3\sqrt{z}}{2}, & 0 \leq z \leq 1 \\ 0, & \text{else} \end{cases} \quad \checkmark$$

6.26 a. $U = Y^m$ so $Y = U^{1/m}$

$$y = h(y) = U^{1/m}$$

$$y = h^{-1}(U^{1/m})$$

$$y = \frac{dh^{-1}(U^{1/m})}{du}$$

$$= \frac{1}{m} U^{(1/m)-1}$$

$$f_U(u) = f_Y(h^{-1}(u)) \left| \frac{dh^{-1}(u)}{du} \right|$$

$$= \left(\frac{m}{2}\right) (u^{1/m})^{m-1} \left(e^{-\frac{1}{2} (u^{1/m})^m}\right) \left(\frac{1}{m}\right) (u^{(1/m)-1})$$

$$= \frac{1}{2} e^{-\frac{u}{2}}$$

$$f_U(u) = \begin{cases} \frac{1}{2} e^{-\frac{u}{2}}, & u > 0 \\ 0, & \text{else} \end{cases} \quad \checkmark$$

6.72 a. $U_1 = \min(Y_1, Y_2)$

$$P(U_1 \leq u_1) = P(\min(Y_1, Y_2) \geq u_1)$$

$$= 1 - P(Y_1 \geq u_1) P(Y_2 \geq u_1)$$

$$= 1 - [1 - F_{Y_1}(u_1)] [1 - F_{Y_2}(u_1)]$$

$$= 1 - [1 - F_Y(u_1)]^2$$

$$= 1 - (1 - u_1)^2$$

cdf $F_{U_1}(u_1) = \begin{cases} 1 - (1 - u_1)^2, & 0 \leq u_1 \leq 1 \\ 0, & \text{else} \end{cases}$

F_U
Now find density
 $f_{U_1}(u_1)$

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Problem 8 a. independent $Y_1, Y_2 \sim \text{Gamma}(\alpha=2, \beta=5)$
 $(Y_1, Y_2) \sim \left(\frac{y_1 e^{-y_1/5}}{25 \Gamma(2)} \right) \left(\frac{y_2 e^{-y_2/5}}{25 \Gamma(2)} \right) = \frac{y_1 y_2 e^{-y_1/5} e^{-y_2/5}}{625 \Gamma(2)}$

$$= \frac{y_1 y_2 e^{-(y_1+y_2)/5}}{625 \Gamma(2)}$$

$$y_1 > 0 \quad y_2 > 0$$

$$y_2 = u_2 - y_1$$

$$y_1 < u_2$$

$$f_{u_2}(u_2) = \int_{y_1=0}^{u_2} \frac{y_1 y_2 e^{-(y_1+y_2)/5}}{625} \bigg| e^{-(u_2-y_1)/5} \bigg| dy_1$$

no u_2 here?

$1/2$

$$= \frac{y_2}{625} e^{-y_2} \int_{y_1=0}^{u_2} y_1 e^{-y_1/5} dy_1$$

$$= \frac{y_2}{625} e^{-y_2} \left(\frac{y_1^2 e^{-y_1/5}}{2} \right) = \frac{y_2 e^{-(u_2+y_2)}}{1250} \quad u_2 > 0$$

$$U \sim \text{Gamma}(\alpha=2, \beta=1250)$$

b. $Y_1, Y_2 \sim \text{Unif}(0,1)$
 $(Y_1, Y_2) \sim \left[\frac{1}{12\pi} e^{-\frac{1}{2}y_1^2} \right] \left[\frac{1}{12\pi} e^{-\frac{1}{2}y_2^2} \right] = \frac{1}{2\pi} e^{-\frac{1}{2}(y_1^2+y_2^2)}$

$$y_1 > 0 \quad y_2 > 0 \quad y_2 = u_2 - y_1 \quad y_1 < u_2$$

$$f_{u_2}(u_2) = \int_{y_1=0}^{u_2} \frac{1}{2\pi} e^{-\frac{1}{2}(y_1^2+y_2^2)} \bigg| e^{-(u_2-y_1)} \bigg| dy_1$$

$$= \frac{1}{2\pi} e^{-\frac{1}{2}y_2^2} \int_{y_1=0}^{u_2} e^{-\frac{1}{2}y_1^2} dy_1$$

$$= \frac{1}{2\pi} e^{-\frac{1}{2}y_2^2} (2 - 2e^{-\frac{1}{2}y_2^2}) \quad u_2 > 0$$

$$U \sim \text{Exp}(\beta=2)$$

uniform,
not normal

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Problem C Bernoulli(p)

$$Y_{(1)} = \min(Y_1, \dots, Y_n)$$

$$P(Y_{(1)} \leq y) = 1 - P(Y_{(1)} > y)$$

$$1 - P(Y_1 > y, Y_2 > y, \dots, Y_n > y)$$

$$1 - P(Y_1 > y)P(Y_2 > y) \dots P(Y_n > y)$$

$$= 1 - [P(Y_1 > y)]^n$$

where did the
n go?

$$Y \sim \begin{cases} 1-p, & y=1 \\ 1-(1-p), & y=0 \end{cases}$$

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$$Y_{(1)} = \max(Y_1, \dots, Y_n)$$

$$P(Y_{(1)} \geq y) = 1 - P(Y_{(1)} < y)$$

$$1 - P(Y_1 < y, Y_2 < y, \dots, Y_n < y)$$

$$1 - P(Y_1 < y)P(Y_2 < y) \dots P(Y_n < y)$$

$$= 1 - [P(Y_1 < y)]^n$$

same?

$$Y \sim \begin{cases} 1-p, & y=1 \\ 1-(1-p), & y=0 \end{cases}$$